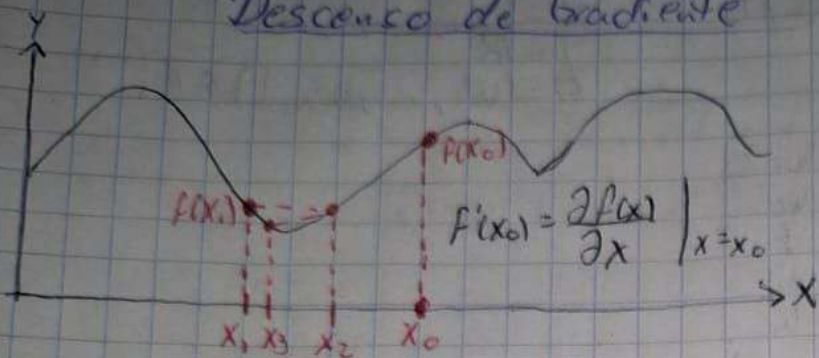


$f(x)$ $f: \mathbb{R} \rightarrow \mathbb{R}$ Forma genérica Para encontrar mínimos

Descenso de Gradiente



η = Paso de aprendizaje
Learning rate

$$\begin{aligned} x_0 & \\ x_1 & \leftarrow x_0 - \eta f'(x_0) \Rightarrow x_{k+1} \leftarrow x_k - \eta f'(x_k) \\ x_2 & \leftarrow x_1 - \eta f'(x_1) \end{aligned}$$

Cuando son varios Parámetros

$$\theta_{k+1} \leftarrow \theta_k - \eta \nabla f(\theta) |_{\theta = \theta_k}$$

$$b_{k+1} \leftarrow b_k - \eta \frac{\nabla f(\theta_k)}{\partial b} E_{in}(w_k, b_k)$$

$$w_{j,k+1} \leftarrow w_{j,k} - \eta \frac{\partial E_{in}(w_k, b_k)}{\partial w_j}$$

H : Conjunto de hipótesis

$$\hat{y} = h_{\theta}(x) = w_1 x_1 + \dots + w_n x_n + b$$

$$x = (x_1, \dots, x_n) \in \mathbb{R}^n$$

$$y \in \mathbb{R}$$

$$\theta = (w_1, \dots, w_n, b) \in \mathbb{R}^{n+1}$$

$$\begin{bmatrix} x_1^{(1)} & \dots & x_n^{(1)} \\ \vdots & & \vdots \\ x_1^{(m)} & \dots & x_n^{(m)} \end{bmatrix} \begin{bmatrix} y^{(1)} \\ \vdots \\ y^{(m)} \end{bmatrix}$$

$X \quad Y$
 $(m, n) \quad (m, 1)$

$$\theta = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \\ b \end{bmatrix}$$

$(n+1, 1)$

Theta que minimice el error

$$\theta^* = \arg \min_{\theta \in \mathbb{R}^{n+1}} \frac{1}{2m} \sum_{i=1}^m (y^{(i)} - [x^{(i)} \ 1] \begin{bmatrix} \psi \\ 1 \end{bmatrix})^2 \quad \text{MSE}$$

$$X_e = \begin{bmatrix} x_1^{(1)} & \dots & x_n^{(1)} & 1 \\ \vdots & & \vdots & \vdots \\ x_1^{(m)} & \dots & x_n^{(m)} & 1 \end{bmatrix}$$

$$\hat{y} = X_e \theta$$

$(m, 1) \quad (m, n+1) \quad (n+1, 1)$
 $(m, 1)$

$$E = Y - \hat{y}$$

$(m, 1) \quad (m, 1) \quad (m, 1)$

$$E: u(\theta) = \frac{1}{2} E^T E$$

$$\theta^* = \underbrace{(X_e^T X_e)^{-1}}_{(n+1, m)} \underbrace{X_e^T}_{(m, n+1)} \underbrace{Y}_{(m, 1)}$$

$(n+1, n+1)$
 $(n+1, 1)$

Recordatorio:

$$h_{\theta}(x) = w_1 x_1 + w_2 x_2 + \dots + w_n x_n = b = w^T x + b = [x^T, 1] \begin{bmatrix} w \\ b \end{bmatrix} = [x^T, 1] \theta$$

$$\theta^* = w^*, b^* = \arg \min_{\theta \in \mathbb{R}^{n+1}} \frac{1}{2M} \sum_{i=1}^M (y^{(i)} - [x^T, 1] \theta)^2$$

Para w_j :

$$\frac{\partial E_{in}(\theta)}{\partial w_j} = \frac{1}{2M} \sum_{i=1}^M -2 (y^{(i)} - \hat{y}^{(i)}) x_j^{(i)}$$

$$\frac{\partial E_{in}(\theta)}{\partial b} = \frac{1}{2M} \sum_{i=1}^M -2 (y^{(i)} - \hat{y}^{(i)})$$

$$\nabla E_{in}(\theta) = \frac{1}{M} \begin{bmatrix} \sum_{i=1}^M (y^{(i)} - \hat{y}^{(i)}) x_1^{(i)} \\ \sum_{i=1}^M (y^{(i)} - \hat{y}^{(i)}) x_2^{(i)} \\ \vdots \\ \sum_{i=1}^M (y^{(i)} - \hat{y}^{(i)}) x_n^{(i)} \\ \sum_{i=1}^M (y^{(i)} - \hat{y}^{(i)}) \end{bmatrix}$$

$$= \frac{1}{M} \begin{bmatrix} \sum_{i=1}^M (y^{(i)} - \hat{y}^{(i)}) x_1^{(i)} \\ \sum_{i=1}^M (y^{(i)} - \hat{y}^{(i)}) x_2^{(i)} \\ \vdots \\ \sum_{i=1}^M (y^{(i)} - \hat{y}^{(i)}) x_n^{(i)} \\ \sum_{i=1}^M (y^{(i)} - \hat{y}^{(i)}) \end{bmatrix}$$

$$= \frac{1}{M} \begin{bmatrix} x_1^{(1)} & x_1^{(2)} & \dots & x_1^{(M)} \\ x_2^{(1)} & x_2^{(2)} & \dots & x_2^{(M)} \\ \vdots & \vdots & \ddots & \vdots \\ x_n^{(1)} & x_n^{(2)} & \dots & x_n^{(M)} \\ 1 & 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} y^{(1)} - \hat{y}^{(1)} \\ y^{(2)} - \hat{y}^{(2)} \\ \vdots \\ y^{(M)} - \hat{y}^{(M)} \end{bmatrix}$$

$$\begin{bmatrix} y^{(1)} - \hat{y}^{(1)} \\ y^{(2)} - \hat{y}^{(2)} \\ \vdots \\ y^{(M)} - \hat{y}^{(M)} \end{bmatrix} = \begin{bmatrix} y^{(1)} - \hat{y}^{(1)} \\ y^{(2)} - \hat{y}^{(2)} \\ \vdots \\ y^{(M)} - \hat{y}^{(M)} \end{bmatrix}$$

$$\theta \leftarrow \theta - \eta \nabla E_{in}(\theta)$$

$$\theta \leftarrow \theta + \frac{\eta}{M} X^T (y - X \theta)$$


```

def dg_lin(X, Y, w0, b0, lr, max_epochs, e_tol):
    M = X.shape[0]
    w = w0.copy()
    b = b0.copy()
    hist = []
    for _ in range(max_epochs):
        Y_est = X @ w + b
        Err = Y - Y_est
        hist.append(np.square(Err).mean())
        grad_w = -(1/M) X.T @ Err
        d_b = Err.mean()
        w -= lr * grad_w
        b -= lr * d_b
        if np.abs(grad_w).max() < e_tol:
            break
    return w, b, hist

```

Cómo llamar la función

X, Y
 $X.shape = [m, n], Y.shape = [m, 1]$
 $w = \text{np.zeros}(X.shape[1])$
 $b = 0$

$w, b, hist = dg_lin(X, Y, w, b, 0.1, 50, 1e-4)$

Descenso de gradiente en Problemas lineales:

$$\begin{aligned}
 \hat{Y} &= Xw + b \\
 Err &= Y - \hat{Y} \\
 E_{in} &= \frac{1}{M} \sum_{i=1}^M Err_i^2 \\
 \nabla_w E_{in} &= -\frac{1}{M} \sum_{i=1}^M X_i^T Err_i \\
 \frac{\partial E_{in}}{\partial b} &= -\frac{1}{M} \sum_{i=1}^M Err_i
 \end{aligned}$$

$$\begin{aligned}
 w &\leftarrow w - lr \nabla_w E_{in} \\
 b &\leftarrow b - lr \frac{\partial E_{in}}{\partial b}
 \end{aligned}$$

$$\theta^* = \arg \min_{\theta} E_{in}(\theta)$$

$$\begin{aligned}
 f \approx h^* &\begin{cases} \nearrow E_{in} \approx 0 \\ \searrow E_{in} \approx E_{out} \end{cases} \\
 \updownarrow &
 \end{aligned}$$

$$E_{out} \approx 0$$

Probablemente Aproximadamente
 correcto
 $P(|E_{out} - E_{in}| \geq \epsilon) \leq \delta$

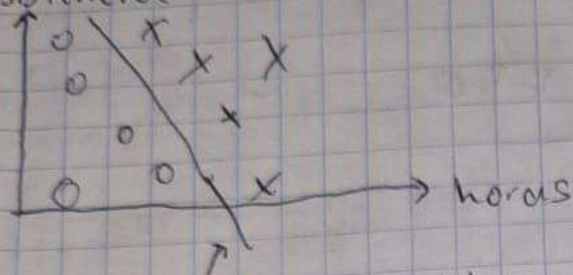
Cuando se cumple
 $E_{in} = E_{out}$

Pero no cumple

$E_{in} \approx 0$

Entonces es un modelo de alto sesgo

asistencia



$$w_1x_1 + w_2x_2 + b = 0$$

M. hipótesis es: $h_\theta = \text{Sign}(w_1x_1 + w_2x_2 + b)$

$$x \in \mathbb{R}^n$$

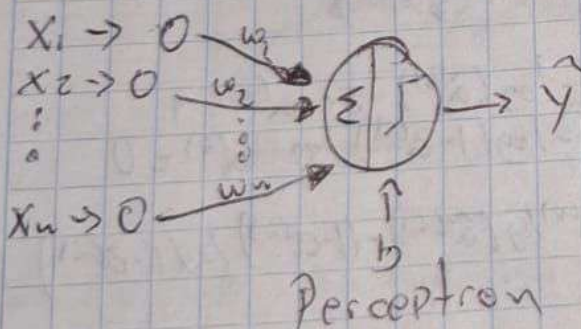
$$y \in \{-1, 1\}$$

$$H = \{h_\theta \mid h_\theta = \text{Sign}(w^T x + b), w \in \mathbb{R}^n, b \in \mathbb{R}, \theta = (w, b)\}$$

$$\text{Loss}(y, \hat{y}) = \begin{cases} 1 & \text{si } y \neq \hat{y} \\ 0 & \text{o.c.} \end{cases} = \max(-y\hat{y}, 0)$$

0/1-loss

$$E_{in}(w, b) = \frac{1}{n} \sum_{i=1}^n \text{loss}(y^{(i)}, \text{Sign}(w^T x^{(i)} + b))$$



Discreto a continuo:

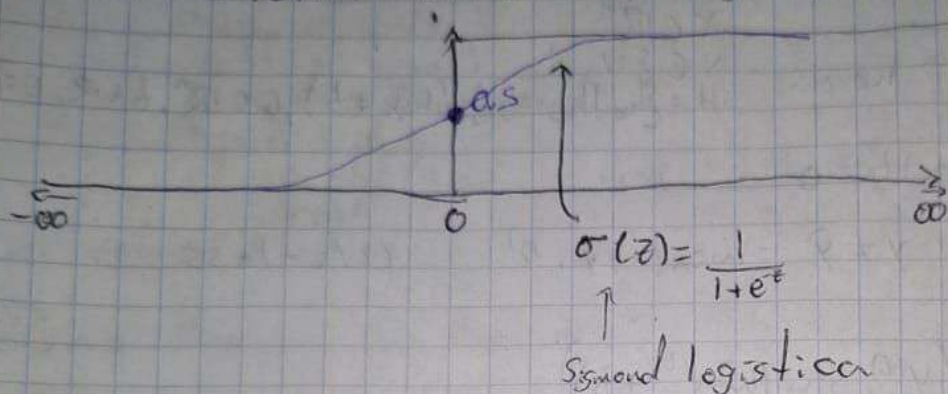
$$\hat{a} = \Pr(y=1 | x=x; \theta)$$

$$\hat{y} = \begin{cases} 1 & \text{si } \hat{a} > 0.5 \\ 0 & \text{o.c.} \end{cases}$$

Regression logistica

$$\hat{a} = \sigma(w_1 x_1 + w_2 x_2 + \dots + w_n x_n + b)$$

$$z = w_1 x_1 + w_2 x_2 + \dots + w_n x_n + b = x^T \theta + b = x^T \theta$$



$$\begin{bmatrix} x_1^{(1)} & \dots & x_n^{(1)} \\ x_1^{(2)} & \dots & x_n^{(2)} \\ \vdots & & \vdots \\ x_1^{(m)} & \dots & x_n^{(m)} \end{bmatrix} \quad \begin{bmatrix} y^{(1)} \\ y^{(2)} \\ \vdots \\ y^{(m)} \end{bmatrix} \quad \begin{bmatrix} a^{(1)} \\ a^{(2)} \\ \vdots \\ a^{(m)} \end{bmatrix}$$

X

Y

A

$$y^{(i)} \in \{-1, 1\}$$

$$E_{in}(w, b) = \frac{1}{M} \sum_{i=1}^M \text{loss}(a^{(i)}, \hat{a}^{(i)})$$

$$\text{loss}(a^{(i)}, \hat{a}^{(i)}) = \begin{cases} -\log(\hat{a}^{(i)}) & \text{si } a^{(i)} = 1 \\ -\log(1 - \hat{a}^{(i)}) & \text{si } a^{(i)} = 0 \end{cases}$$

$$\text{loss}(a^{(i)}, \hat{a}^{(i)}) = -a^{(i)} \log(\hat{a}^{(i)}) - (1 - a^{(i)}) \log(1 - \hat{a}^{(i)})$$

Descenso de gradiente:

$$w \leftarrow w - \eta \frac{\partial}{\partial w} E_{in}(w, b)$$

$$b \leftarrow b - \eta \frac{\partial}{\partial b} E_{in}(w, b)$$

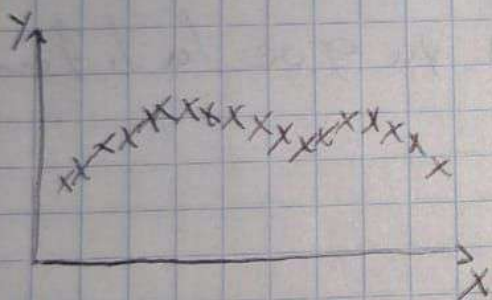
$$\frac{\partial}{\partial w_j} E_{in}(w, b) = \frac{\partial}{\partial w_j} \frac{1}{M} \sum_{i=1}^M -a^{(i)} \log(\hat{a}^{(i)}) - (1 - a^{(i)}) \log(1 - \hat{a}^{(i)})$$

$$\text{donde } \hat{a}^{(i)} = \frac{1}{1 + e^{-z^{(i)}}}, \text{ y } z^{(i)} = w_1 x_1^{(i)} + \dots + w_n x_n^{(i)} + b$$

$$= \frac{1}{M} \sum_{i=1}^M -\frac{a^{(i)}}{\hat{a}^{(i)}} + \frac{1 - a^{(i)}}{1 - \hat{a}^{(i)}} - \frac{\partial a^{(i)}}{\partial w_j}$$

$$\begin{aligned}
\frac{\partial \tilde{a}}{\partial w_j} &= \frac{\partial}{\partial w_j} \frac{1}{1 + e^{-z^{(i)}}} \\
&= \frac{\partial}{\partial z^{(i)}} (1 + e^{-z^{(i)}})^{-1} \frac{\partial z^{(i)}}{\partial w_j} \\
&= (1 + e^{-z^{(i)}})^{-2} e^{-z^{(i)}} x_j^{(i)} \\
&= \left(\frac{1 + e^{-z^{(i)}}}{(1 + e^{-z^{(i)}})^2} - \frac{1}{(1 + e^{-z^{(i)}})^2} \right) x_j^{(i)} \\
&= \left(\frac{1}{1 + e^{-z^{(i)}}} - \frac{1}{1 + e^{z^{(i)}}} \right) x_j^{(i)} = (\tilde{a}^{(i)} - \tilde{a}^{(i)^2}) x_j^{(i)} \\
&= \tilde{a}^{(i)} (1 - \tilde{a}^{(i)}) x_j^{(i)}
\end{aligned}$$

Si tenemos un conjunto de datos y queremos aplicar regresión.



Creamos una función:

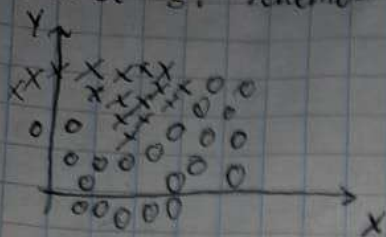
$$\phi: \mathbb{R}^n \rightarrow \mathbb{R}^n$$

$$x \in \mathbb{R}$$

$$x' = \phi(x) \equiv (x, x^2, x^3, x^4)$$

$$x' = \phi(x) \equiv (\phi_1(x), \phi_2(x), \dots, \phi_n(x))$$

Ahora si tenemos algo como lo siguiente



Entonces vamos a querer sacar la expansión Polinomial

$$x = (x_1, x_2)$$

$$\phi(x) = (x_1, x_2, x_1^2, x_1 x_2, x_2^2)$$

$$\phi(x) = (x_1, x_2, x_1^2, x_1 x_2, x_2^2, x_1^3, x_1^2 x_2, x_1 x_2^2, x_2^3)$$

$$x = (x_1, x_2, x_3, x_4)$$

$$\phi(x) = (x_1, x_2, x_3, x_4, x_1^2, x_1 x_2, x_1 x_3, x_1 x_4, x_2^2, x_2 x_3, x_2 x_4, x_3^2, x_3 x_4, x_4^2)$$

La dimension bc se saca contando todos los parámetros libres

$$h(x) = w^T \phi(x) + b \quad w, \phi(x) \in \mathbb{R}^{n'}, K \in \mathbb{R}$$

Aquí, la dimension bc es igual a $n'+1$ ya que la b también se cuenta como parámetro libre.

$$\|w\|_2 = \sqrt{w^T w} \rightarrow \text{Norma } L_2$$

$$\|w\|_1 = \sum_{j=1}^n |w_j| \rightarrow \text{Norma } L_1$$

$$h(x) = w^T \phi(x) + b \quad w, \phi(x) \in \mathbb{R}^{n'}, K \in \mathbb{R}$$

$$\text{bajo } \underbrace{\sum_{j=1}^n w_j^2 \leq C}_{\leftarrow \text{Regularización}}$$